Stable phase retrieval in function spaces

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Based on joint works with M. Christ and B. Pineau, as well as with D. Freeman, T. Oikhberg and B. Pineau

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Can we *stably* recover f from the above measurements?

Stable phase retrieval

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Note that if $f, g \in E$ and $|f| = |g|$, then f is a multiple of g. This implication is known as *phase retrieval*.

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Phase retrieval says that the map $|f| \mapsto f/\sim$ is well-defined on E; stable phase retrieval additionally requires this map to be Lipschitz.

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By Plancherel $\inf_{|\lambda|=1} \|F - \lambda G\|_{L_2} = \inf_{|\lambda|=1} \|\widehat{F} - \lambda \widehat{G}\|_{L_2}$.

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Notice that we have eliminated the operator from the inequality and encoded it into the subspace.

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By the above argument, T does stable phase retrieval if and only if $T(H) \subseteq L_2(\mu)$ does stable phase retrieval in our sense.

Goal of the talk

Problem:

Cahill, Casazza, Daubechies, Trans. Amer. Math. Soc., 2016

Alaifari, Grohs, SIAM J. Math. Anal., 2017

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Give simple and natural examples of subspaces of $L_p(\mu)$ doing *stable* phase retrieval, both over $\mathbb{R}$ and $\mathbb{C}$.

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Give simple and natural examples of subspaces of $L_p(\mu)$ doing *stable* phase retrieval, both over $\mathbb{R}$ and $\mathbb{C}$. Give the first examples doing $\mathcal{F}$-SPR.

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Reminder on definitions

A subspace $E \subseteq L_p(\mu)$ does *C-stable phase retrieval* if for all $f, g \in E$ we have

$$\inf_{|\lambda|=1} \|f - \lambda g\|_{L_p} \leq C \||f| - |g|\|_{L_p}.$$

Part I: Obstructions and examples

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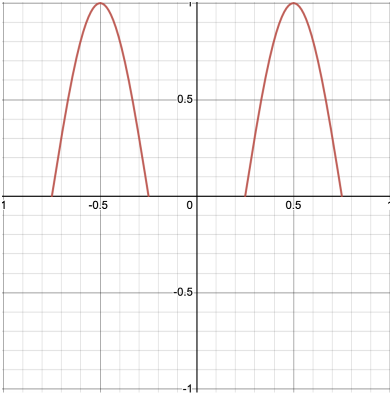

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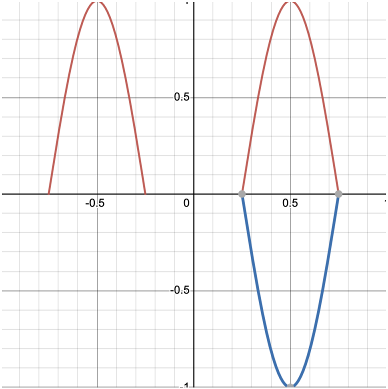

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For *real* scalars, this is the only obstruction to stable phase retrieval

Theorem (Freeman, Oikhberg, Pineau, T.)

A subspace E of a real Banach lattice does C-stable phase retrieval if and only if E contains no $\frac{1}{C}$-almost disjoint pairs.

Examples - real scalars

Theorem (Christ, Pineau, T.)

$\overline{span}\{\sin(2\pi 4^n x) : n \in \mathbb{N}\} \subseteq L_p([0,1];\mathbb{R})$ *does stable phase retrieval for all* $1 \leq p < \infty$.

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Theorem (Christ, Pineau, T.)

$\overline{span}\{r_n\} \subseteq L_4(\mu;\mathbb{R})$ *of mean zero iid random variables does SPR iff* $|r_n|$ *is not identically constant.*

Additional constraint in the complex case

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In the complex case, an SPR subspace $E \subseteq L_2(\mu)$ cannot contain linearly independent real vectors.

If $f, g \in E$ are real and linearly independent then $f + ig$ and $f - ig$ have the same modulus, but are not linearly dependent.

Examples - complex scalars

Theorem (Christ, Pineau, T.)

Let $P \in L_p([0,1];\mathbb{C})$, $1 \leq p < \infty$, *be of the form*

$$P(x) = \sum_{k=1}^{N} a_k e^{2\pi i k x}, \ \ a_k \in \mathbb{C}.$$

If $|P|$ *is not constant and* $A > 2N$, $\overline{\mathit{span}}\{P(A^n x) : n \in \mathbb{N}\}$ *does SPR.*

Part II: Proof that complex polynomials do SPR.

Idea of the proof

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Note $|r_n|^2 - 1 = 0$ is an enemy in phase retrieval.

Idea: Use this decomposition to avoid this.

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Hence, orthonormal sequences with appropriate orthogonality and lower bounds on their products do SPR in L_4.

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Theorem (Christ, Pineau, T.)

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The fact that $|P|$ is not constant and $r_n(\cdot) = P(A^n \cdot)$ is a dilate of a single function gives us a uniform lower bound on $|r_n|^2 - 1$.

Sparseness of the dilates give non-overlapping Fourier supports on all required products, hence the required orthogonality.

No negative frequencies implies no possible cancellations leading to non-overlapping Fourier support.

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Question: How do we get to L_p? How do we get "denser" examples?

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In particular, if E does SPR in L_p, then we may view E as a closed subspace of L_q for $1 \leq q \leq p$.

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Our SPR examples: $e^{2\pi ix} \rightsquigarrow P(x)$.

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Connection to $\Lambda(p)$-sets

A subset $\Lambda \subseteq \mathbb{Z}$ is called $\Lambda(p)$ if $\operatorname{span}\{e^{2\pi inx} : \; n \in \Lambda\}$ is $\bigstar_p$.

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Open for all other p!

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Recall that we have constructed $E \subseteq L^4(\mu)$ doing SPR and we know

$$\text{SPR in } L_p \Rightarrow \bigstar_p \Rightarrow \bigstar_q \quad (q < p).$$

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Moreover, by Hölder we have

$$\inf_{|\lambda|=1} \|f - \lambda g\|_{L_2} \leq \||f| - |g|\|_{L_2}^{\gamma} (\|f\|_{L_2} + \|g\|_{L_2})^{1-\gamma}.$$

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In other words, the map $|f| \mapsto f/\sim$ is well-defined and Hölder continuous on the unit ball of E in the L_2 norm.

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More precisely, if $|f| \mapsto f/\sim$ is well-defined and Hölder continuous on the unit ball of E then it is automatically Lipschitz continuous on E.

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Theorem (Freeman, Oikhberg, Pineau, T.)

Let $f, g \in L_2(\mu)$. Then there exists $f', g' \in span\{f, g\}$ such that $f' \perp g'$ and

$$\||f'| - |g'|\| \leq \||f| - |g|\|,$$

$$\inf_{|\lambda|=1} \|f - \lambda g\| \leq \inf_{|\lambda|=1} \|f' - \lambda g'\|.$$

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In L_p, suffices to check SPR on "well-separated" vectors.

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Answer: If $p > 2$, yes.

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Is the extra integrability necessary?

Answer: If $p > 2$, yes. If $p \leq 2$, no!

References

1. M. Christ, B. Pineau, and T., *Examples of Hölder-stable phase retrieval*, Math. Res. Lett., to appear.
2. D. Freeman, T. Oikhberg, B. Pineau, and T., *Stable phase retrieval in function spaces*, Math. Ann., to appear.

Many open questions!

1. STFT and uncertainty principles.
2. Problems from quantum mechanics: Have more information, want stronger conclusions.
3. Declipping and packing problems.